1. N-Queens Dry Run (N = 4)

Initial State

board =

....

....

....

....

Step-by-step Execution

Step 1: row = 0

Try col = 0 → place Q

Q...

....

....

....

Step 2: row = 1

col = 0 ✗ (same column)

col = 1 ✗ (diagonal)

col = 2 ✓ → place Q

Q...

..Q.

....

....

Step 3: row = 2

col = 0 ✗

col = 1

X

col = 2 X

col = 3 X

No position → BACKTRACK

Step 4: Backtrack to row = 1

remove 2 from (1,2)

try col = 3 ✓

2...

...2

....

....

Step 5: row = 2

col = 0 X

col = 1 ✓ → place 2

2...

...2

.2..

....

Step 6: row = 3

all columns X

👉 BACKTRACK

again

Eventually, valid solution found:

. 2 . .

. . 2

2 . . .

. .2 .

✓ Final Output

Total solutions = 2